

TOP 40 SQL INTERVIEW QUESTIONS

Question 1 of 40

Topic: SELECT Statement

Definition: SELECT Statement is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 1: Select all rows from a table.

Input

employees

Program

```
SELECT * FROM employees;
```

Output

All employee rows

TOP 40 SQL INTERVIEW QUESTIONS

Question 2 of 40

Topic: Column Selection

Definition: Column Selection is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 2: Select specific columns.

Input

```
name, salary
```

Program

```
SELECT name, salary FROM employees;
```

Output

```
Name and salary columns
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 3 of 40

Topic: WHERE Clause

Definition: WHERE Clause is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 3: Filter rows using WHERE.

Input

```
salary > 50000
```

Program

```
SELECT * FROM employees WHERE salary > 50000;
```

Output

```
Employees earning above 50000
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 4 of 40

Topic: AND Operator

Definition: AND Operator is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 4: Use AND condition.

Input

```
department='IT' and salary>60000
```

Program

```
SELECT * FROM employees WHERE department = 'IT' AND salary >
60000;
```

Output

```
Matching IT employees
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 5 of 40

Topic: OR Operator

Definition: OR Operator is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 5: Use OR condition.

Input

```
department='IT' or 'HR'
```

Program

```
SELECT * FROM employees WHERE department IN ('IT', 'HR');
```

Output

```
IT and HR employees
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 6 of 40

Topic: Ascending Sort

Definition: Ascending Sort is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 6: Sort records ascending.

Input

salary

Program

```
SELECT * FROM employees ORDER BY salary ASC;
```

Output

Employees sorted by salary

TOP 40 SQL INTERVIEW QUESTIONS

Question 7 of 40

Topic: Descending Sort

Definition: Descending Sort is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 7: Sort records descending.

Input

salary

Program

```
SELECT * FROM employees ORDER BY salary DESC;
```

Output

Employees sorted descending

TOP 40 SQL INTERVIEW QUESTIONS

Question 8 of 40

Topic: DISTINCT Clause

Definition: DISTINCT Clause is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 8: Find unique departments.

Input

department

Program

```
SELECT DISTINCT department FROM employees;
```

Output

Unique departments

TOP 40 SQL INTERVIEW QUESTIONS

Question 9 of 40

Topic: COUNT Function

Definition: COUNT Function is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 9: Count rows.

Input

```
employees
```

Program

```
SELECT COUNT(*) FROM employees;
```

Output

```
Total row count
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 10 of 40

Topic: AVG Function

Definition: AVG Function is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 10: Find average salary.

Input

salary

Program

```
SELECT AVG(salary) FROM employees;
```

Output

Average salary

TOP 40 SQL INTERVIEW QUESTIONS

Question 11 of 40

Topic: MAX Function

Definition: MAX Function is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 11: Find maximum salary.

Input

salary

Program

```
SELECT MAX(salary) FROM employees;
```

Output

Highest salary

TOP 40 SQL INTERVIEW QUESTIONS

Question 12 of 40

Topic: MIN Function

Definition: MIN Function is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 12: Find minimum salary.

Input

salary

Program

```
SELECT MIN(salary) FROM employees;
```

Output

Lowest salary

TOP 40 SQL INTERVIEW QUESTIONS

Question 13 of 40

Topic: SUM Function

Definition: SUM Function is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 13: Find total salary.

Input

salary

Program

```
SELECT SUM(salary) FROM employees;
```

Output

Total salary

TOP 40 SQL INTERVIEW QUESTIONS

Question 14 of 40

Topic: GROUP BY Clause

Definition: GROUP BY Clause is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 14: Group by department.

Input

```
department, salary
```

Program

```
SELECT department, AVG(salary) FROM employees GROUP BY  
department;
```

Output

```
Average salary by department
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 15 of 40

Topic: HAVING Clause

Definition: HAVING Clause is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 15: Filter groups using HAVING.

Input

```
average salary > 60000
```

Program

```
SELECT department, AVG(salary) AS avg_salary FROM employees GROUP  
BY department HAVING AVG(salary) > 60000;
```

Output

```
Departments above threshold
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 16 of 40

Topic: LIKE Operator

Definition: LIKE Operator is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 16: Use LIKE for pattern matching.

Input

names beginning with A

Program

```
SELECT * FROM employees WHERE name LIKE 'A%';
```

Output

Names beginning with A

TOP 40 SQL INTERVIEW QUESTIONS

Question 17 of 40

Topic: BETWEEN Operator

Definition: BETWEEN Operator is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 17: Use BETWEEN.

Input

```
salary 40000 to 70000
```

Program

```
SELECT * FROM employees WHERE salary BETWEEN 40000 AND 70000;
```

Output

```
Employees in salary range
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 18 of 40

Topic: IS NULL Condition

Definition: IS NULL Condition is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 18: Use IS NULL.

Input

```
manager_id
```

Program

```
SELECT * FROM employees WHERE manager_id IS NULL;
```

Output

```
Employees without manager
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 19 of 40

Topic: CASE Expression

Definition: CASE Expression is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 19: Use CASE.

Input

salary category

Program

```
SELECT name, CASE WHEN salary >= 60000 THEN 'High' ELSE 'Normal'  
END AS category FROM employees;
```

Output

Salary category per employee

TOP 40 SQL INTERVIEW QUESTIONS

Question 20 of 40

Topic: INNER JOIN

Definition: INNER JOIN is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 20: Use INNER JOIN.

Input

employees and departments

Program

```
SELECT e.name, d.department_name FROM employees e INNER JOIN  
departments d ON e.department_id=d.department_id;
```

Output

Matching employee and department rows

TOP 40 SQL INTERVIEW QUESTIONS

Question 21 of 40

Topic: LEFT JOIN

Definition: LEFT JOIN is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 21: Use LEFT JOIN.

Input

employees and departments

Program

```
SELECT e.name, d.department_name FROM employees e LEFT JOIN  
departments d ON e.department_id=d.department_id;
```

Output

All employees with matching departments

TOP 40 SQL INTERVIEW QUESTIONS

Question 22 of 40

Topic: RIGHT JOIN

Definition: RIGHT JOIN is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 22: Use RIGHT JOIN.

Input

employees and departments

Program

```
SELECT e.name, d.department_name FROM employees e RIGHT JOIN  
departments d ON e.department_id=d.department_id;
```

Output

All departments with matches

TOP 40 SQL INTERVIEW QUESTIONS

Question 23 of 40

Topic: Duplicate Detection

Definition: Duplicate Detection is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 23: Find duplicate emails.

Input

email

Program

```
SELECT email, COUNT(*) FROM employees GROUP BY email HAVING  
COUNT(*) > 1;
```

Output

Duplicate email values

TOP 40 SQL INTERVIEW QUESTIONS

Question 24 of 40

Topic: Second Highest Salary

Definition: Second Highest Salary is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 24: Find second highest salary.

Input

salary

Program

```
SELECT MAX(salary) FROM employees WHERE salary < (SELECT  
MAX(salary) FROM employees);
```

Output

Second highest salary

TOP 40 SQL INTERVIEW QUESTIONS

Question 25 of 40

Topic: Top-N Query

Definition: Top-N Query is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 25: Find top 3 salaries.

Input

salary

Program

```
SELECT DISTINCT salary FROM employees ORDER BY salary DESC LIMIT  
3;
```

Output

Top three salary values

TOP 40 SQL INTERVIEW QUESTIONS

Question 26 of 40

Topic: Subquery

Definition: Subquery is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 26: Use a subquery.

Input

salary above average

Program

```
SELECT * FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
```

Output

Employees above average

TOP 40 SQL INTERVIEW QUESTIONS

Question 27 of 40

Topic: EXISTS Operator

Definition: EXISTS Operator is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 27: Use EXISTS.

Input

orders for customers

Program

```
SELECT * FROM customers c WHERE EXISTS (SELECT 1 FROM orders o
WHERE o.customer_id=c.customer_id);
```

Output

Customers with orders

TOP 40 SQL INTERVIEW QUESTIONS

Question 28 of 40

Topic: UNION Operator

Definition: UNION Operator is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 28: Use UNION.

Input

two employee lists

Program

```
SELECT name FROM current_employees UNION SELECT name FROM  
former_employees;
```

Output

Combined unique names

TOP 40 SQL INTERVIEW QUESTIONS

Question 29 of 40

Topic: Anti-Join Query

Definition: Anti-Join Query is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 29: Find records in one table but not another.

Input

customers without orders

Program

```
SELECT c.* FROM customers c LEFT JOIN orders o ON  
c.customer_id=o.customer_id WHERE o.customer_id IS NULL;
```

Output

Customers without orders

TOP 40 SQL INTERVIEW QUESTIONS

Question 30 of 40

Topic: CREATE TABLE

Definition: CREATE TABLE is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 30: Create a table.

Input

employees

Program

```
CREATE TABLE employees (id INT PRIMARY KEY, name VARCHAR(100),  
salary DECIMAL(10,2));
```

Output

Table created

TOP 40 SQL INTERVIEW QUESTIONS

Question 31 of 40

Topic: INSERT Statement

Definition: INSERT Statement is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 31: Insert a row.

Input

```
id=1, name='Asha'
```

Program

```
INSERT INTO employees (id, name, salary) VALUES (1, 'Asha',  
55000);
```

Output

```
One row inserted
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 32 of 40

Topic: UPDATE Statement

Definition: UPDATE Statement is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 32: Update a row.

Input

```
id=1 salary=60000
```

Program

```
UPDATE employees SET salary = 60000 WHERE id = 1;
```

Output

```
Salary updated
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 33 of 40

Topic: DELETE Statement

Definition: DELETE Statement is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 33: Delete a row.

Input

```
id=1
```

Program

```
DELETE FROM employees WHERE id = 1;
```

Output

```
Row deleted
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 34 of 40

Topic: ALTER TABLE

Definition: ALTER TABLE is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 34: Add a column.

Input

phone

Program

```
ALTER TABLE employees ADD phone VARCHAR(20);
```

Output

phone column added

TOP 40 SQL INTERVIEW QUESTIONS

Question 35 of 40

Topic: COALESCE Function

Definition: COALESCE Function is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 35: Use COALESCE.

Input

missing manager

Program

```
SELECT name, COALESCE(manager_id, 0) AS manager_id FROM  
employees;
```

Output

NULL manager IDs become 0

TOP 40 SQL INTERVIEW QUESTIONS

Question 36 of 40

Topic: RANK Window Function

Definition: RANK Window Function is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 36: Rank employees by salary.

Input

salary

Program

```
SELECT name, salary, RANK() OVER (ORDER BY salary DESC) AS  
salary_rank FROM employees;
```

Output

Salary ranking

TOP 40 SQL INTERVIEW QUESTIONS

Question 37 of 40

Topic: Running Total

Definition: Running Total is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 37: Calculate running total.

Input

```
salary ordered by id
```

Program

```
SELECT id, salary, SUM(salary) OVER (ORDER BY id) AS  
running_total FROM employees;
```

Output

```
Running salary total
```

TOP 40 SQL INTERVIEW QUESTIONS

Question 38 of 40

Topic: Monthly Sales Aggregation

Definition: Monthly Sales Aggregation is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 38: Find monthly sales.

Input

order_date and amount

Program

```
SELECT EXTRACT(MONTH FROM order_date) AS month, SUM(amount) FROM
orders GROUP BY EXTRACT(MONTH FROM order_date);
```

Output

Sales total by month

TOP 40 SQL INTERVIEW QUESTIONS

Question 39 of 40

Topic: Latest Record

Definition: Latest Record is a SQL technique used to retrieve, filter, transform, or summarize database data. It helps write clear queries for the required database operation.

Question 39: Find the latest record.

Input

```
order_date
```

Program

```
SELECT * FROM orders ORDER BY order_date DESC LIMIT 1;
```

Output

```
Latest order
```